

Q) Define the following.

- i) Slack and Surplus Variable
- ii) BFS
- iii) G.N. L.P.P
- iv) Dummy Activity
- v) PERT , CPM and RAMP

Ans: In **Linear Programming (LP)** and **Optimization**, **slack** and **surplus** variables are used to convert inequalities into equalities. This is necessary when applying methods like the **Simplex Method**, which require equations in standard form.

◆ **1. Slack Variable:** A **slack variable** is **added** to a “less than or equal to” ($\leq$) constraint to convert it into an equation.

✓ **Example:**

$$x + y \leq 10$$

Add a slack variable $s \geq 0$:

$$x + y + s = 10$$

◆ Here, s represents the **unused (slack)** part of the resource.

- If $s = 0 \Rightarrow$ the resource is fully used.
- If $s > 0 \Rightarrow$ some resource is left unused.

◆ **Surplus Variable:** A **surplus variable** is **subtracted** from a “greater than or equal to” ($\geq$) constraint to convert it into an equation.

✓ **Example:**

$$x + y \geq 10$$

Subtract a surplus variable $s \geq 0$:

$$x + y - s = 10$$

◆ Here, s represents the **excess (surplus)** over the minimum required.

- If $s = 0 \Rightarrow$ the minimum requirement is exactly met.
- If $s > 0 \Rightarrow$ there's an excess over the minimum.

ii) BFS: In the context of **Linear Programming (LPP)**, a **Basic Feasible Solution** is a solution to a system of linear equations that:

1. **Satisfies all the constraints** (i.e., it's feasible), and
2. Is a **basic solution**, meaning it corresponds to a solution obtained by setting some variables (called non-basic variables) to zero and solving for the remaining (basic) variables.

- A **basic solution** is obtained by selecting any mmm variables (if there are mmm equations), solving the equations for them, and setting the remaining $n - mn - mn - m$ variables to zero.

iii) G.N. LPP: A **Linear Programming Problem (LPP)** is an optimization technique used to find the maximum or minimum value of a **linear objective function**, subject to a set of **linear constraints** (equalities or inequalities).

Example:

Maximize $Z = 3x + 5y$

Subject to:

$2x + y \leq 100$

$X + 3Y \leq 90$

$X, Y \geq 0$

iv) Dummy Activity: An artificial or imaginary activity that shows **logical dependencies** between tasks, but **does not consume any time or resources**.

◆ Purpose of Dummy Activity

Dummy activities are used to:

1. **Preserve the correct logical relationship** between activities.
2. **Avoid confusion** when two activities:
 - Have the **same start and end nodes**, but
 - Are dependent on different activities.
3. Make the **network diagram accurate and unambiguous**.

◆ Example

Suppose:

- **Activity A** and **Activity B** both start from the same event.
- **Activity C** depends on **A** only.
- **Activity D** depends on **both A and B**.

To represent this:

- A **dummy activity** is used from the end of A to a new node before C, to separate dependencies.

Network Representation:

A Dummy C

1 -----> 2 -----> 3 ----->

B D

1 -----> 4 -----> 5

Here, the dummy activity (dashed arrow from 2 to 3) shows that C depends only on A, not B.

V) PERT, CPM and RAMP : PERT: The **program evaluation and review technique (PERT)** is a statistical tool used in [project management](#), which was designed to analyze and represent the [tasks](#) involved in completing a given [project](#).

CPM: CPM is used to determine the earliest possible starting time for each task in the project. It'll also help a project manager determine the critical tasks and non-critical tasks. This helps when executing the project. If you're behind schedule, you know which tasks can be skipped without impacting the integrity of the work.

RAMP: Resource Allocation and Multi-Project Scheduling, or RAMPS, involves managing resources like software, hardware, and other hard costs to appropriately meet an organization needs. Employees and human capital are also considered resources that must be divided to accomplish required tasks. A number of factors play into RAMPS such as balancing competing priorities, determining the most effective course of action to maximize limited resources, and making returns on investments.

Q) Difference Between Assignment and Transportation Problem

Parameters	Transportation Problem	Assignment Problem
Definition	It deals with the optimal distribution of goods or resources from multiple sources (factories and warehouses) to multiple destinations (retail stores or customers).	It is a special type of transportation problem that assigns n origins (workers or instances) to n destinations (jobs or machines).
Objective	Minimize the total transportation cost	Minimize total cost or time associated with completing tasks
Constraints	1. Supply Constraints (The amount supplied from each source) 2. Demand Constraints (Total amount demanded at each destination).	1. Each task is assigned to exactly one resource. 2. Each resource is assigned to exactly one task.
Types	Balanced and Unbalanced	Balanced and Unbalanced
Method	Northwest, Least Cost, Vogels Approximation, or Modi Method. Transportation methods can be used to solve assignment problems.	Hungarian Method Assignment methods can't be used to solve transportation problems.
Real-World Application	Supply Chain Optimization, Distribution Network Design, Traffic Flow Optimization	Job Scheduling Personnel, Assignment Facility Location, Machine Allocation

Ans: Assignment Problem: An Assignment Problem is a special type of Transportation Problem in Operational Research that deals with assigning n origins (workers or instances) to n destinations (jobs or machines). The goal of the assignment problem is to determine the minimum cost of the assignment. Transportation problem: A transportation problem is a Linear Programming Problem that deals with identifying an optimal solution for transportation and allocating resources to various destinations and from one site to another while keeping the expenditure to a minimum.

